

Andy Liu

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Professional Experience

Fastenal – Software Developer

August 2023 – Present

- Developed and maintained enterprise applications using Java 8 and 21, adhering to strict security and coding standards.
- Refactored a monolithic Terraform architecture into modular roots for core cloud infrastructure and application tiers across Dev, Test, Stage, and Prod, enforcing state isolation and zero-downtime lifecycle protection.
- Provisioned and managed Azure infrastructure using Terraform, creating modules for virtual machines, Key Vaults, Event Hubs, Azure Container Apps, Private Endpoints, and Managed Identities.
- Migrated legacy Tomcat/EAP background workloads to Spring Boot microservices, configuring engine heartbeats, scheduled jobs, and asynchronous background processing.
- Built and integrated RESTful APIs with Spring Boot and Oracle SQL databases.
- Designed and implemented Azure DevOps CI/CD pipelines for Java and Flutter applications, improving deployment efficiency and reliability.
- Automated database migrations with Liquibase, implementing preview and validation stages in Azure DevOps pipelines with Managed Identity authentication.
- Built and deployed Docker images from Azure Container Registry (ACR) to containers and hosted local containers for testing.
- Replaced shell-dependent archive extraction mechanisms with native Java implementations to streamline release artifact processing.
- Designed a custom AI-assisted code-review agent and structured prompting workflows to analyze pull requests, evaluate Terraform execution plans, and enforce criteria validation for Java and Terraform codebases.
- Diagnosed multi-environment production issues, network/TLS certificate failures, and container runtime errors to ensure application stability.

Revature – Software Developer

July 2022 – January 2023

- Developed full-stack applications using Angular and Spring Boot.
- Deployed scalable solutions on AWS.
- Managed and optimized PostgreSQL and H2 databases to ensure data efficiency and integrity.
- Completed tasks on schedule as part of Agile teams.

Education

University of Tennessee, Knoxville

Graduated: May 2020

Bachelor of Science in Computer Science

GPA: 3.6

Relevant Coursework: Algorithms, Database Management, Software Engineering

Technical Skills

- **Languages:** Java (8, 21), Python, C++, JavaScript (TypeScript, JS), SQL, Shell Scripting (Bash)
- **Frameworks & Architecture:** Spring Boot, Microservices, Angular, Apache Tomcat, RESTful APIs, Flutter, Maven, Quartz Scheduler
- **Cloud & Infrastructure:** Azure (Container Apps, Container Registry, Storage, Key Vault, Managed Identities, Virtual Networks, Private Endpoints, Private DNS, Log Analytics), AWS, Terraform (IaC)
- **DevOps & CI/CD:** Azure DevOps Pipelines (YAML), CI/CD Automation, Docker, Liquibase, TLS/Certificates
- **AI & Developer Tools:** AI-Assisted Code Reviews (Java & Terraform), Prompt Engineering for AI Coding, GitHub Copilot, VS Code Agentic AI, Git, Jira
- **Databases:** Oracle SQL, PostgreSQL, MySQL, H2
- **Methodologies:** Agile, Kanban, AI-Assisted Development